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United States Patent	12385774
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Cuniasse; Pierre-Antoine et al.

Weighing device with a wheatstone bridge assembly and method

Abstract

A weighing device of the electronic bathroom scale type, includes four feet, each fitted with a load cell, each load cell including two resistors combined in a Wheatstone bridge type of assembly to define a first axis of symmetry and a second axis of symmetry, the assembly further including at least one auxiliary circuit configured to selectively short-circuit two resistors belonging to the same foot or to two neighboring feet in the bathroom scale, the first auxiliary circuit being symmetrical relative to the first axis or the second axis, the assembly being controlled by an electronic control unit to estimate a weight and to estimate an offset of a user on the bathroom scale.

Inventors:	Cuniasse; Pierre-Antoine (Issy les Moulineaux, FR), Faussard; Guillaume (Issy les Moulineaux, FR), Premel; Xavier (Issy les Moulineaux, FR)
Applicant:	WITHINGS (Issy les Moulineaux, FR)
Family ID:	71452469
Assignee:	WITHINGS (Issy les Moulineaux, FR)
Appl. No.:	17/794220
Filed (or PCT Filed):	April 15, 2021
PCT No.:	PCT/FR2021/050661
PCT Pub. No.:	WO2021/209720
PCT Pub. Date:	October 21, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230138345 A1	May. 04, 2023

Foreign Application Priority Data

FR

2003832

Apr. 16, 2020

Publication Classification

Int. Cl.: G01G3/14 (20060101); G01G19/44 (20060101)

U.S. Cl.:

CPC G01G3/1406 (20130101); G01G19/44 (20130101);

Field of Classification Search

CPC: G01G (19/44); G01G (3/1406)

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Primary Examiner: Huls; Natalie

Attorney, Agent or Firm: Pillsbury Winthrop Shaw Pittman LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. National Stage of PCT/FR2021/050661, filed Apr. 15, 2021, which in turn claims priority to French patent application number 2003832 filed Apr. 16, 2020. The content of these applications are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(2) This disclosure relates to weighing devices and methods. More particularly, this disclosure relates to the field of electronic bathroom scales.

PRIOR ART

(3) Electronic bathroom scales, also called personal scales, are commonly equipped with four feet, each of these feet comprising a load cell. Such a configuration makes it possible to measure the weight of a user when he stands on the scale.

(4) Conventionally, the load cells of the scale are arranged in a Wheatstone bridge type of assembly. Then the loads measured by each of the load cells can be used to estimate the weight of a user.

(5) However, it has been found that when the loads measured by each of the load cells are not equal, the estimate of the user's weight can vary. This variability is due in particular to imperfections in the assembly used, as well as to the user being offset relative to the center of the scale.

(6) Thus, document WO2014/013208 describes an optimized load cell assembly, for establishing the weight of a user as well as for measuring his offset relative to the center of the scale. The assembly has two separate Wheatstone bridges, each comprising two load cells connected to a dedicated amplifier. The measured offset is transmitted to the user, allowing the user to recenter himself in order to improve the weight estimate.

(7) This assembly allows effectively estimating an offset of the user. However, this assembly requires two Wheatstone bridges, connected to two amplifiers, which leads to an increase in energy consumption related to weighing. In addition, it has been found that, depending on the arrangement of the load cells in the Wheatstone bridges, the estimation of an offset in the front/rear direction or the right/left direction remains imprecise and with room for improvement. In addition, this assembly requires intervention by the user, who must lean or reposition himself to correct his offset.

(8) There is therefore a need for a scale which makes it possible to reduce the influence of the offset of a user, which does not have the disadvantages of the prior art.

(9) Moreover, there are devices, for example Nintendo's Wii Balance Board, which exploit the user's offset to interact with a platform, for example to play games. However, this type of device uses four Wheatstone bridges, resulting in a high production cost.

SUMMARY

(10) A weighing device of the electronic bathroom scale type is proposed that comprises four feet, respectively left front, right front, left rear, and right rear, the left front foot comprising a left front load cell, the right front foot comprising a right front load cell, the left rear foot comprising a left rear load cell, and the right rear foot comprising a right rear load cell,

(11) each load cell comprising at least two resistors,

(12) the load cells being combined in a Wheatstone bridge type of assembly comprising:

(13) a first branch and a second branch, the first and second branches being mounted in parallel between a reference voltage and a ground potential, the first and second branches being arranged on either side of a first axis of symmetry passing through the reference voltage and the ground potential; a first intermediate point, on the first branch, and a second intermediate point, on the second branch, the first and second branch comprising the same number of resistors on either side of the first and second intermediate point, respectively, in order to define a second axis of symmetry passing through the first and second intermediate points; and a first auxiliary circuit, configured to selectively short-circuit two resistors belonging to the same foot or to two neighboring feet (called the first foot (feet)) in the bathroom scale, the first auxiliary circuit being symmetrical relative to the first axis or the second axis,

the assembly being controlled by an electronic control unit, to estimate a weight and to estimate an offset of a user on the bathroom scale.

(14) Advantageously, such an assembly makes it possible to modify the contribution of a load cell to a measurement of the output signal from the assembly. Indeed, through several measurements of the output signal with different contributions, it appears possible to ascertain an estimate of the weight and the offset of a user on the scale. Thus, it appears possible to estimate the weight of the user cleared of any offset, without any intervention by the user. In addition, it appears possible to exploit the offset dynamically to allow the user to interact with the scale, in order to clean a ballistocardiography (BCG) signal and/or to analyze the user's balance. In addition, the proposed assembly reduces the components and energy required for the bathroom scale to operate.

(15) The features set forth in the following paragraphs may optionally be implemented. They may be implemented independently of each other or in combination with each other.

(16) The assembly may comprise a second auxiliary circuit, the second auxiliary circuit being configured to short-circuit two resistors belonging to the same second foot or to two neighboring second feet in the bathroom scale, the second auxiliary circuit being symmetrical relative to the first axis or second axis. In particular, at least one resistor short-circuited by the first auxiliary circuit and at least one resistor short-circuited by the second auxiliary circuit belong to two neighboring feet.

(17) Thus, by the use of two auxiliary circuits, three measurements of the output signal from the assembly can be made, each with contributions from different load cells. It is therefore possible to ascertain a value algebraically for the weight and offset of the user on the scale in the front/rear direction and in the right/left direction.

(18) The assembly may comprise a third auxiliary circuit, the third auxiliary circuit being symmetrical relative to the first axis or second axis, the first, second, and third auxiliary circuits being configured to concern three among the four load cells. This means that the first, second, and third auxiliary circuits are configured to short-circuit three of the four load cells or to short-circuit at least one resistor of three load cells among the four load cells.

(19) Thus, the third auxiliary circuit can allow a fourth measurement of the output signal from the assembly, each of the measurements having contributions from different load cells. In addition to the weight and the offsets, it is then possible to ascertain a torsion value for the plate on which the load cells of the bathroom scale are mounted.

(20) The set of resistors affected by the short circuits of the auxiliary circuit(s) can belong to at least two separate load cells.

(21) The or each auxiliary circuit may include at least one switch, for example a transistor, controlled by the electronic control unit. The switch can also be an opto-isolator.

(22) Thus, the or each auxiliary circuit can be selectively controlled to an open state or a closed state, to form a short circuit. Then several measurements of the output signal can be made.

(23) The or each auxiliary circuit may have two switches, preferably two transistors, controlled by the electronic control unit.

(24) A transistor is not perfectly symmetrical; this configuration allows better symmetry of the assembly.

(25) The or each auxiliary circuit can be connected to at least one among: the reference voltage, the ground potential, the first intermediate point, or the second intermediate point. When the auxiliary circuit is connected to the reference voltage or to the ground potential, a reduction in the signal-to-noise ratio (SNR) is observed, while the rest of the bridge is powered at the reference voltage during the short circuit.

(26) The intermediate points can be connected to a single amplifier in order to obtain an output signal from the assembly.

(27) This configuration makes it possible to obtain an output signal usable by the electronic control unit to estimate the users weight and offset. In addition, the single amplifier allows better energy efficiency of the device, requiring only one amplifier to obtain a usable output signal.

(28) The device may comprise a display, configured to display information to a user. The user can

thus be informed of the measurements made by the scale, in particular his weight. The display also allows user interaction with the scale.

(29) The device may comprise a communication module configured to exchange data with a smartphone and/or a server. The user can thus access the measurements made by the bathroom scale, at a distance from the bathroom scale. Measurement tracking can be carried out. The personal scale can also receive additional data from the server and/or the smartphone, for example about the user on the personal scale.

(30) According to another aspect, a method implemented in the weighing device is proposed, comprising: performing a first measurement of the output signal from the assembly; controlling the first auxiliary circuit into the closed state; performing a second measurement of the output signal from the assembly; establishing a weight and offset of a user based on the first and second measurements.

(31) This method makes it possible to perform measurements of the output signal from the assembly with contributions from different load cells. The output signal of each measurement can be used by the bathroom scale to establish the user's weight and offset. The weight can be transmitted to the user, and the offset can be used to clean the BCG signal, analyze a user's balance, and/or allow the user to interact with the bathroom scale.

(32) The features set forth in the following paragraphs may optionally be implemented. They may be implemented independently of each other or in combination with each other.

(33) The method may include, after the second measurement: controlling the second auxiliary circuit into the closed state; performing a third measurement of the output signal from the assembly; establishing a weight, front-rear offset, and right-left offset of a user, based on the first, second, and third measurements.

(34) The method may include, after the second measurement: controlling the second auxiliary circuit into the closed state; performing a third measurement of the output signal from the assembly; establishing a weight of a user and the bending of a plate of the weighing device, based on the first, second, and third measurements.

(35) With the third measurement, it is possible to ascertain a value algebraically for the weight and offset of the user on the scale in the front/rear direction and in the right/left direction.

(36) The method may comprise, after the third measurement: controlling the third auxiliary circuit into the closed state; performing a fourth measurement of the output signal from the assembly; establishing the weight, front-rear offset, right-left offset of a user and the torsion of a plate of the weighing device, based on the first, second, third, and fourth measurements.

(37) With the fourth measurement, it is possible to estimate the torsion of the plate which occurs during the measurements. Estimating the torsion can improve the precision of the weight and offsets. Estimating the torsion can also reduce the requirements for plate rigidity without impacting the precision of the weight and offsets.

(38) The weight, front-rear offset, and right-left offset can be calculated by multiplying a vector, composed of the first, second, and third measurements, by a transition matrix.

(39) The transition matrix can be obtained by a theoretical calculation.

(40) The transition matrix can be obtained experimentally. This allows improving the precision of the transition matrix.

(41) The weight may be displayed to a user.

(42) The front-rear and/or right-left offset can allow at least one among: an interaction of the bathroom scale with a user, a cleaning of a ballistocardiography (BCG) signal, and/or an analysis of the user's balance on the bathroom scale.

(43) The order of the measurements is of no particular importance.

(44) The method described may further comprise a step of controlling an electronic system, using the offset information obtained. In this manner, the weighing device can be used as a controller of an electronic system. The electronic system may comprise the weighing device itself, a (separate)

computer, or a game console.

(45) The measurements made by the bathroom scale can be sent to a smartphone and/or a server. The user can thus access the measurements made by the bathroom scale, at a distance from the bathroom scale. Measurement tracking can be carried out.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Other features, details and advantages will become apparent upon reading the detailed description below, and upon analyzing the accompanying drawings, in which:

(2) FIG. 1 is a general view of the weighing device according to the invention, viewed from above;

(3) FIG. 2a shows a first display on a display of the weighing device of FIG. 1;

(4) FIG. 2b shows a second display on the display of the weighing device of FIG. 1;

(5) FIG. 2c shows a third display on the display of the weighing device of FIG. 1;

(6) FIG. 3 shows a first embodiment of the basic wiring diagram of the weighing device of FIG. 1, in a first mode of operation;

(7) FIG. 4 shows the wiring diagram of FIG. 3 in a second mode of operation;

(8) FIG. 5 shows the wiring diagram of FIG. 3 in a third mode of operation;

(9) FIG. 6 shows a variant of the basic wiring diagram of the weighing device of FIG. 3;

(10) FIG. 7 shows another variant of the basic wiring diagram of the weighing device of FIG. 3;

(11) FIG. 8A shows other variants of the basic wiring diagram of the weighing device of FIG. 3;

(12) FIG. 8B shows other variants of the basic wiring diagram of the weighing device of FIG. 3;

(13) FIG. 9 shows a second embodiment of the basic wiring diagram of the weighing device of FIG. 1;

(14) FIG. 10 shows variants of the basic wiring diagram of the weighing device of FIG. 9;

(15) FIG. 11 shows a third embodiment of the basic wiring diagram of the weighing device of FIG. 1;

(16) FIG. 12 shows a fourth embodiment of the basic wiring diagram of the weighing device of FIG. 1;

(17) FIG. 13 shows a flowchart of the method implemented by the weighing device of FIG. 1;

(18) FIG. 14 shows an example of a matrix calculation for obtaining the weight and the offsets;

(19) FIG. 15A shows a fifth embodiment of the basic wiring diagram of the weighing device of FIG. 1;

(20) FIG. 15B shows a variant of the wiring diagram illustrated in FIG. 15;

(21) FIG. 16 schematically shows a perspective view of a sub-assembly of the weighing device of FIG. 1;

(22) FIG. 17 shows a flowchart of the method implemented by the weighing device of FIG. 1.

DESCRIPTION OF EMBODIMENTS

(23) In the various figures, the same references designate identical or similar elements.

(24) 1. General Shape of the Bathroom Scale

(25) FIG. 1 shows an example of a weighing device 10 according to one embodiment of the invention.

(26) In the illustrated example, this weighing device 10 is presented as an electronic bathroom scale, or personal scale, on which a user can position himself in order to measure his weight in particular.

(27) The electronic bathroom scale 10 comprises a main body of generally rectangular or square shape and four feet respectively arranged near the four corners of the body, each foot comprising measurement means.

(28) Alternatively, the scale could have a round or oval shape. In this case, the four feet are evenly

distributed around the center axis of the bathroom scale, each foot comprising the measurement means.

(29) Specifically, the left front foot comprises a left front load cell **3AG**, the right front foot comprises a right front load cell **3AD**, the left rear foot comprises a left rear load cell **3PG**, and the right rear foot comprises a right rear load cell **3PD**.

(30) Note that, as can be seen in FIG. **16**, the aforementioned load cells are arranged on a plate **58** provided in the electronic bathroom scale **10**. Each load cell extends between a point of contact **60** with the plate **58**, and the ground GND. Thus, each load cell performs a measurement relative to the main plane of extension of the plate **58**. The plate **58** directly takes the weight of the user. In one embodiment, the plate **58** is the plate the user positions himself upon.

(31) In FIG. **1**, the front-rear direction is referenced by the X axis, the letter A designating the front and the letter P designating the rear, while the right-left direction is referenced by the Y axis, extending between the left side denoted G and the right side denoted D.

(32) With regard to the load cells already mentioned, load cells comprising two strain gauges of known type are preferably chosen, in particular gauges comprising a first element whose resistance increases under the effect of vertical compression applied to the foot in question and a second element whose resistance decreases under the effect of said vertical compression.

(33) In the example illustrated here, the right front load cell **3AD** comprises such a first element called the first right front resistor **15**, and such a second element called the second right front resistor **16**.

(34) Similarly, the right rear load cell **3PD** comprises such a first element called the first right rear resistor **17**, and such a second element called the second right rear resistor **18**.

(35) Likewise for the left side, the left front load cell **3AG** comprises such a first element called the first left front resistor **19**, and such a second element called the second left front resistor **20**.

(36) Finally, the left rear load cell **3PG** comprises such a first element called the first left rear resistor **21**, and such a second element called the second left rear resistor **22**.

(37) “Neighbors” means two feet or load cells that share a “front/rear” or “left/right” characteristic. For example, the neighbors of load cell **3AG** are the load cells **3AD** and **3PG**. In an architecture with four feet or load cells, only the feet or load cells diagonally across from each other are not neighbors.

(38) The electronic bathroom scale **10** further comprises an electronic control unit **12** and a display **14**.

(39) As shown in FIGS. **2a** to **2c**, the display **14** can display information **24**, for example the estimated weight for the object or the name of the user present on the bathroom scale **10**. In addition, the display **14** may display one or more indicators **26**. The function of these indicators is to allow the user to interact with the bathroom scale according to a positional offset relative to an ideal centered position in which the forces are distributed substantially evenly over the four feet of the bathroom scale.

(40) In the example of FIG. **2a**, the display **14** displays the estimated weight **24** for the object or user present on the bathroom scale **10**. The display shows four indicators **26**, in the form of arrows. The indicators can indicate to the user on the bathroom scale the direction to lean in order to achieve the ideal centered position.

(41) In the example of FIG. **2b**, the display shows two options C1, C2. The options may for example be two different users, or two answers to a previously displayed question. The display shows an indicator **26**, to allow the user to select among the two options via an offset in the right/left direction. The user will be able to lean to the right or to the left to change his selection.

(42) In the example of FIG. **2c**, the display shows a list of options C1, C2, C3. The list can consist of various functionalities of the scale, for example displaying the weather forecast, displaying a weighing history, or accessing the settings. The list may also be a plurality of possible answers to a question previously displayed.

(43) The display then includes “forward” **26a** and “back” **26b** indicators, to allow the user to navigate through the options. The user will be able to lean forward to move up the list, and backward to move down.

(44) The display also includes “right” **26c** and “left” **26d** indicators to allow the user to select an option or return to the list. The user will be able to lean to the right to select an option, or to the left to go back.

(45) For each of the examples described above, an indicator can be activated to indicate to the user the direction to follow in order to correct the observed offset, or to navigate and select among the options. Activation of the indicator may consist of making it flash, turning it on while the others remain off, moving it within the display, or any other manner.

(46) Of course, other types of displays which allow displaying information and interacting with the bathroom scale according to the offset can be implemented.

(47) **2. Wheatstone Bridge**

(48) Referring to FIG. **3**, a basic wiring diagram of the bathroom scale comprises a Wheatstone bridge type of assembly **28**, which combines the resistors **15** to **22** of the four load cells.

(49) Each of the resistors **15-22** respectively has a resistance value denoted R1-R8. In the example illustrated here, the odd resistors, or resistors with positive sensitivity (hereinafter positive resistors), increase with the vertical compressive force applied to the feet, while conversely the even resistors, or resistors with negative sensitivity (hereinafter negative resistors), decrease with the force applied.

(50) In a typical example, resistors with the same nominal value are chosen, for example 500 ohms or 1 kilohm (1 k Ω).

(51) In addition, odd resistors define a negative pole A of the load cell, and even resistors define a positive pole B. A third pole C separates the even and odd resistors of each load cell. In other words, pole C is the midpoint of the load cell.

(52) In the Wheatstone bridge type of assembly **28**, the negative poles A of the even resistors are joined together and the negative poles B of the odd resistors are joined together to form a quadrilateral having ends at the four poles C of the load cells.

(53) Furthermore, two opposite poles C in the quadrilateral are respectively connected to a reference voltage Vs and a ground potential GND. Thus, the Wheatstone bridge type of assembly **28** defines a first branch **30** and a second branch **32**, mounted in parallel between the reference voltage Vs and the ground potential GND.

(54) The reference voltage Vs comes from a voltage source stabilized at a constant predetermined value, for example 2.8 V in the example considered. Alternatively, the assembly could also be powered by a source of current.

(55) In the example illustrated, the first branch **30** connects, in series: the reference voltage Vs to pole C of the right front load cell **3AD**; the negative pole A of the right front load cell **3AD** to the negative pole A of the left front load cell **3AG**; the positive pole B of the left front load cell **3AG** to the positive pole B of the right rear load cell **3PD**; and pole C of the right rear load cell **3PD** to the ground potential.

(56) Similarly, the second branch **32** connects, in series: the reference voltage Vs to pole C of the right front load cell **3AD**; the positive pole B of the right front load cell **3AD** to the positive pole B of the left rear load cell **3PG**; the negative pole A of the left rear load cell **3PG** to the negative pole A of the right rear load cell **3PD**; and pole C of the right rear load cell **3PD** to the ground potential.

(57) The two remaining poles C, not connected to the reference voltage or to the ground potential, define intermediate points **34** and **36**. The respective voltages present at the first and second intermediate points are measured, paying particular attention to the potential difference between these two intermediate points, as will be seen below.

(58) As illustrated, the first and second intermediate points **34**, **36** are defined by the poles C of the left front **3AG** and left rear **3PG** load cells, respectively.

(59) Note that the load cells could be mounted differently than the example shown. A pole C of any load cell could be connected to the reference voltage, to the ground potential, or could form one of the intermediate points. Nevertheless, the following constraints will be satisfied: each of the two branches **30**, **32** has the same number of positive resistors and negative resistors; the intermediate points **34**, **36** separate the negative resistors and the positive resistors, of each of the branches.

(60) It thus appears possible to define a first axis of symmetry **A1** and a second axis of symmetry **A2**.

(61) The first axis of symmetry **A1** passes through the connection point of the reference voltage and the connection point of the ground potential and comprises the first branch **30** on the one hand and the second branch **32** on the other hand.

(62) The second axis of symmetry **A2** passes through the first and second intermediate points **34**, **36** and comprises the negative resistors of the first branch **30** and the positive resistors of the second branch **32** on the one hand, and the positive resistors of the first branch **30** and the negative resistors of the second branch **32** on the other hand.

(63) The first and second intermediate points **34**, **36** are connected to an amplifier **38**, whose role is to amplify the potential difference between the two intermediate points, and to deliver these amplified values from an output **40** connected to the control unit **12**.

(64) The voltage read from the first intermediate point is denoted E_{d1} , the voltage read from the second intermediate point is denoted E_{d2} , the voltage read at the output of the amplifier **38**, which is the output signal, is denoted ΔV . We then have the equation $\Delta V = G(E_{d1} - E_{d2})$, G being the gain of the amplifier.

(65) The assembly described above then makes it possible to mount the four load cells in a single Wheatstone bridge, and only requires one amplifier. The number of assembly components and the energy required to perform a measurement of an output signal are reduced, particularly in comparison to document WO2014/013208 where two Wheatstone bridges are associated with two amplifiers.

(66) 3. Auxiliary Circuits

(67) Furthermore, the Wheatstone bridge type of assembly **28** comprises a first auxiliary circuit **42**. As illustrated, the first auxiliary circuit **42** is connected to the negative pole A of the right front load cell **3AD** and to the positive pole B of the right rear load cell **3AD**.

(68) One will note that the first auxiliary circuit **42** is symmetrical relative to the first axis of symmetry **A1**. The assembly is thus balanced. This prevents saturation of the amplifier **38**.

(69) The first auxiliary circuit **42** here comprises a transistor **T1**. When transistor **T1** is open (also called "at rest" or "OFF"), the impedance of the auxiliary circuit is infinite, and no current flows through the auxiliary circuit. The output signal ΔV is not influenced by the first auxiliary circuit.

(70) On the other hand, when transistor **T1** is closed, it becomes equivalent to a wire. Then, as can be seen in FIG. 4, a short circuit forms between the positive B and negative A poles of the right front load cell **3AD**. Then, the assembly comprises a short circuit which cancels out the contribution of resistors **15** and **16** of the right front load cell **3AD**. The resistors do not contribute to the output signal ΔV . The rest of the bridge is supplied with $\frac{3}{4}$ of the reference voltage V_s .

(71) Of course, other types of switches could be considered for controlling the auxiliary circuit or auxiliary circuits described. For example, the switch can be an opto-isolator. The use of an opto-isolator makes it possible to decouple the electrical measurement circuit from the control circuit, and therefore makes it possible to avoid parasitic effects related to the control circuit.

(72) In addition, the Wheatstone bridge type of assembly **28** here comprises a second auxiliary circuit **44**. As illustrated, the second auxiliary circuit is connected to the positive pole B of the right rear load cell **3PD**, and to the negative pole A of the right rear load cell **3PD**.

(73) One will note that, in the same manner as the first auxiliary circuit **42**, the second auxiliary circuit **44** is symmetrical relative to the first axis of symmetry **A1**.

(74) The second auxiliary circuit **44** also includes a transistor **T2**. As with the first transistor **T1**,

when transistor T2 is open, the output signal ΔV is not influenced by the second auxiliary circuit **44**. When transistor T2 is closed, and as illustrated in FIG. 5, a short circuit is formed between the positive B and negative A poles of the right rear load cell **3AD**, and the resistors **17** and **18** of the right rear load cell **3PD** do not contribute to the output signal ΔV . The rest of the bridge is supplied with $\frac{3}{4}$ of the reference voltage V_s .

(75) Each of the two auxiliary circuits **42**, **44** makes it possible to short-circuit the two resistors of load cells belonging to two neighboring feet in the bathroom scale and to cancel out their contribution to the output signal ΔV . In fact, in FIG. 3, the load cells concerned are the two load cells on the right side of the bathroom scale **10**. The auxiliary circuits can thus make it possible to ascertain values corresponding to the weight, to the offset in the right/left direction, and to the offset in the front/rear direction of the user on the bathroom scale.

(76) To do this, the electronic control unit **12** independently controls the activation or deactivation of transistors T1 and T2 by control lines **46** and **48**.

(77) The electronic control unit **12** can then perform a first measurement M1 of the output signal ΔV with the two transistors in the open state. All load cells contribute substantially equally to the output signal ΔV . Measurement M1 is then a weight measurement influenced by the user's right/left and front/rear offsets on the scale.

(78) Furthermore, as illustrated in FIG. 4, the control unit **12** can perform a second measurement M2 of the output signal ΔV with the first transistor T1 in the closed state. The right front load cell **3AD** no longer contributes to the output signal ΔV . Measurement M2 is then a weight measurement particularly influenced by the left and rear load cells.

(79) Finally, as illustrated in FIG. 5, the control unit **12** can perform a third measurement M3 of the output signal ΔV with the second transistor T2 in the closed state. The right rear load cell **3PD** no longer contributes to the output signal ΔV . Measurement M3 is then a weight measurement particularly influenced by the left and front load cells.

(80) Each of the measurements M1, M2, M3 corresponds to linear combinations of the weight and imbalances in the right/left and front/rear direction of the user on the scale. In the control unit **12**, the vector formed by measurements M1, M2, M3 can then be multiplied by a transition matrix A to obtain values corresponding to the weight P, to the front/rear offset A/P, and to the right/left offset D/G.

(81) As can be seen in FIG. 14, the matrix A is formed by coefficients A1-A9. The coefficients can be multiplied by measurements M1, M2, and M3. The value of each of the coefficients can be determined by a theoretical calculation. However, the coefficients can also be determined experimentally, to improve the precision of the transition matrix A. In practice, the experimental matrix A can be viewed as the sum of the theoretical matrix A and a disruption matrix.

(82) The weight P can be transmitted to the display **14**. The displayed weight is then cleared of disruptions due to an offset of the user on the bathroom scale.

(83) In addition, a ballistocardiography (BCG) signal, measured by the scale, can be cleaned of components related to movements of the user in the right/left and front/rear direction. The BCG signal is therefore improved.

(84) The offset values can also allow the user to interact with the bathroom scale **10**, in particular by means of the display **14**.

(85) The offset values can also be used to study the imbalance of the user on the scale.

(86) 4. Variants of Auxiliary Circuits

(87) FIG. 6 also illustrates an assembly **28** comprising two auxiliary circuits **42**, **44**. Here, the first auxiliary circuit **42** comprises a pair of transistors T1, T3 controlled simultaneously by control line **46**. Likewise, the second auxiliary circuit **44** comprises a pair of transistors T2, T4 controlled simultaneously by control line **48**. As a transistor is not perfectly symmetrical, the use of two transistors per auxiliary circuit allows better symmetry of the assembly **28**.

(88) Note that one among the first auxiliary circuit **42** and the second auxiliary circuit **44** could

comprise two transistors and the other among the first auxiliary circuit **42** and the second auxiliary circuit **44** could comprise one transistor.

(89) FIG. 7 illustrates two auxiliary circuits **42**, **44**, each comprising a pair of transistors T1, T3 and T2, T4. Furthermore, an additional branch **50** connects the first auxiliary circuit **42**, between two transistors T1, T3, to the reference voltage V_s . Another additional branch **52** connects the second auxiliary circuit **44** to the ground potential. The additional branches **50**, **52** make it possible to power the bridge assembly at the reference voltage V_s when the transistors of one or the other among the auxiliary circuits are controlled into the closed state. This configuration then makes it possible to increase the value of the output signal ΔV , in particular enough to improve the signal-to-noise ratio (SNR).

(90) The table of FIG. 8A illustrates alternative positions of the auxiliary circuits **42**, **44**. Each variant can comprise auxiliary circuits with one or two transistors, or comprise an additional branch.

(91) Variant V1 corresponds to the assembly described above. Advantageously, the arrangement of the first and second auxiliary circuits **42**, **44** on the load cells linked to the reference voltage and to the ground potential makes it possible to limit the effects of temperature.

(92) In variant V2, the first auxiliary circuit **42** makes it possible to short-circuit the left front load cell **3AG**, for the second measurement M2, and the second auxiliary circuit makes it possible to short-circuit the right front load cell **3AD**, for the third measurement M3.

(93) Variant V3 comprises a first auxiliary circuit **42** making it possible to short-circuit the right rear load cell **3PD**, for the second measurement M2, and a second auxiliary circuit **44** making it possible to short-circuit the left rear load cell **3PG**, for the third measurement M3.

(94) Variant V4 comprises a first auxiliary circuit **42** making it possible to short-circuit the left front load cell **3AG**, for the second measurement M2, and a second auxiliary circuit **44** making it possible to short-circuit the left rear load cell **3PG**, for the third measurement M3.

(95) Note that in each of the variants V1, V2, V3, V4: each auxiliary circuit is symmetrical relative to axis A1 or A2, to avoid amplifier saturation; the first auxiliary circuits and second auxiliary circuits are arranged on the resistors of load cells belonging to two neighboring feet in the bathroom scale.

By satisfying these conditions, it appears possible to perform three measurements of the output signal ΔV to ascertain the weight P of a user, a front/rear offset A/P, and a right/left offset R/L.

(96) In two other variants illustrated in the table of FIG. 8B, the first auxiliary circuits **42** and second auxiliary circuits **44** are arranged on the resistors of load cells belonging to two feet which are not neighboring, meaning two diagonally opposite feet. There are then two possible variants Vd1, Vd2: the left front **3AG** and right rear **3DP** load cells are short-circuited and the right front **3AD** and left rear **3PG** load cells are short-circuited.

(97) In variant Vd1, the first auxiliary circuit **42** makes it possible to short-circuit the right front load cell **3AD**, for the second measurement M2, and the second auxiliary circuit **44** makes it possible to short-circuit the left rear load cell **3PG**, for the third measurement M3.

(98) In variant Vd2, the first auxiliary circuit **42** makes it possible to short-circuit the left front load cell **3AG**, for the second measurement M2, and the second auxiliary circuit **44** makes it possible to short-circuit the right rear load cell **3PD**, for the third measurement M3.

(99) Note that in each of the variants Vd1, Vd2, each auxiliary circuit **42**, **44** is symmetrical relative to axis A1 or A2, to avoid amplifier saturation. By satisfying these conditions, it appears possible to carry out three measurements of the output signal ΔV to obtain the weight and information about torsion of the plate. More information about the torsion will be given below.

(100) 5. Offloaded Auxiliary Circuit

(101) FIG. 9 illustrates an auxiliary circuit **42** comprising two parts **42a**, **42b**. As illustrated, the first part **42a** is connected to the negative pole A of the left front load cell **3AG** and to pole C of the left front load cell **3AG**. The second part **42b** is connected to the positive pole B of the left rear

load cell 3PG and to pole C of the left rear load cell 3PG.

(102) Note that this two-part auxiliary circuit **42** is symmetrical relative to axis **A1**. In addition, note that the first part **42a** and the second part **42b** of the auxiliary circuit **42** short-circuit two resistors belonging to load cells of two neighboring feet in the bathroom scale, in this case the left front and left rear feet.

(103) Each part **42a**, **42b** of the auxiliary circuit **42** comprises a transistor **T1**, **T5**. The two transistors are simultaneously controlled into the closed state by the control unit **12** via control line **44**.

(104) In this closed state, resistor **20** of left front load cell **3AG** and resistor **21** of left rear load cell **3PG** are each short-circuited across their terminals. The influence of the two load cells at the left side of the bathroom scale on the output signal ΔV , during the second measurement **M2**, is then halved. Such a measurement **M2** then makes it possible to ascertain an offset in the right/left direction.

(105) The table in FIG. **10** illustrates variants of two-part auxiliary circuits **42** allowing the second measurement **M2** to be carried out.

(106) Variant **V5** corresponds to the example in FIG. **9**.

(107) Variant **V6** allows short-circuiting resistor **19** of the left front load cell **3AG** and short-circuiting resistor **22** of the left rear load cell **3PG**.

(108) Variant **V7** allows short-circuiting resistor **R2** of the right front load cell **3AD** and resistor **R3** of the right rear load cell **3PD**.

(109) Finally, variant **V8** allows short-circuiting resistor **15** of the right front load cell **3AD** and resistor **18** of the right rear load cell **3PD**.

(110) Note that each of variants **V5** to **V8** of the two-part auxiliary circuit **42** is symmetrical relative to axis **A1** or axis **A2**. Furthermore, each part of the auxiliary circuit is designed to short-circuit two resistors belonging to two neighboring feet in the bathroom scale **10**.

(111) Also note that variants **V5** to **V8** all allow performing a measurement **M2** which is particularly sensitive to an offset in the right/left direction.

(112) Indeed, the arrangement of the two-part auxiliary circuit is limited by the condition of symmetry relative to the first or second axis of symmetry **A1**, **A2**. In order to arrange a two-part auxiliary circuit on a front resistor and on a rear resistor, it is necessary to swap the arrangement of the load cells in the assembly **28**.

(113) **6. Second Auxiliary Circuit of an Offloaded Circuit**

(114) In order to perform a third measurement **M3** on a two-part auxiliary circuit, it is possible to add a second auxiliary circuit **44** in any one of variants **V5** to **V8**. It then appears possible to estimate the weight **P** of a user, a front/rear offset, and a right/left offset.

(115) As can be seen in FIG. **11**, the first auxiliary circuit **42** is analogous to variant **V5** described above. The second auxiliary circuit **44** is connected to the negative pole **A** of the right front load cell **3AD** on the one hand, and to the positive pole **B** of the right front load cell **3AD** on the other hand. The second auxiliary circuit **44** is analogous to the first auxiliary circuit of variant **V1** described above.

(116) Note that the second auxiliary circuit is symmetrical relative to axis **A1**. In addition, the right front load cell **3AD**, short-circuited by the second auxiliary circuit **44**, is neighboring the left front **3AG** and left rear **3PG** load cells in the bathroom scale, which are (partially) short-circuited by the first two-part auxiliary circuit **42**.

(117) Alternatively, the second auxiliary circuit **44** can be any of the auxiliary circuits of variants **V1** to **V4**. Note that in each of the possible combinations, at least one resistor short-circuited by the first auxiliary circuit **42** belongs to a load cell neighboring a resistor short-circuited by the second auxiliary circuit **44**.

(118) Furthermore, as illustrated in the example of FIG. **12**, the second auxiliary circuit **44** can be in two parts, of which one part is shared with the first auxiliary circuit **42**.

(119) Here, the first auxiliary circuit **42** is analogous to variant V5 described above. The second auxiliary circuit comprises a first part **44a** in common with the second part **42b** of the first auxiliary circuit **42**. The second part **44b** of the second auxiliary circuit is connected to the negative pole A of the right rear load cell **3PD** and to pole C of the left rear load cell **3PG**, in order to short-circuit resistor **22**.

(120) Note that the second auxiliary circuit **44** is then symmetrical relative to axis A2. Also note that resistor **20**, short-circuited by the first auxiliary circuit, belongs to the left front foot neighboring the left rear foot operated by the second auxiliary circuit **44**.

(121) Each part of the auxiliary circuit is controlled independently by a transistor T1, T5, T6. The second measurement M2 is carried out when the control unit **12** simultaneously closes transistors T1 and T5 of the first and second auxiliary circuits **42**, **44**. Then, measurement M2 is analogous to the two-part circuit of variant V5 described above. The third measurement M3 is performed by simultaneously controlling transistors T5 and T6 of the second and third auxiliary circuits **44**, **54**. Measurement M3 is then analogous to the third measurement M3 of variant V4.

(122) It appears possible to conceive of other arrangements of the second auxiliary circuit **44**, not shown. In order to ascertain the weight and the right/left and front/rear offsets, the condition must nevertheless be satisfied that at least one resistor short-circuited by the first auxiliary circuit and at least one resistor short-circuited by the second auxiliary circuit belong to two neighboring feet in the bathroom scale.

(123) 7. Supplemental Auxiliary Circuit

(124) As can be seen in FIG. 15A, the Wheatstone bridge type of assembly **28** can also include a third auxiliary circuit **56**. In the example illustrated, the third auxiliary circuit **56** (with a transistor T7) is connected between the negative pole A and the positive pole B of the left front load cell **3AG**.

(125) Note that here the third auxiliary circuit **56** is symmetrical relative to the axis of symmetry A2. The assembly **28** remains balanced. Indeed, the auxiliary circuits **42**, **44**, **56** each have symmetry relative to axis A1 or axis A2.

(126) Similarly to the first and second auxiliary circuits **42**, **44**, the third auxiliary circuit **56** may include one or two transistors with or without an additional branch. Then, when the transistor(s) is/are open, the output signal ΔV is not influenced by the third auxiliary circuit **56**. Conversely, when the transistor(s) is/are closed, a short circuit forms between the positive and negative poles of the left front load cell **3AG**. Then, the assembly **28** comprises a short circuit which cancels out the contribution of resistors **19** and **20** of the left front load cell **3AG**.

(127) The first, second, and third auxiliary circuits **42**, **44**, **56** make it possible to short-circuit three of the four load cells of the bathroom scale **10**. In the example of FIG. 15A, the three load cells concerned are the right and left front load cells **3AD**, **3AG** and the right rear load cell **3PD**. The three auxiliary circuits **42**, **44**, **56** allow the introduction of a fourth measurement dimension. The fourth dimension can be used to ascertain the weight, the front/rear offset, the right/left offset, and a torsion value of the plate **58** on which the load cells are mounted.

(128) FIG. 15B illustrates an alternative to the embodiment of FIG. 15A, in which one of the auxiliary circuits is offloaded (meaning in two parts which each short-circuit a resistor of a different load cell), as described in relation to FIG. 9. The circuit of FIG. 15a corresponds to that of FIG. 12, with the addition of a third auxiliary **56** which is identical to the auxiliary circuit **44** of FIGS. 2 to 5. Other variants combining three auxiliary circuits of which at least one is offloaded are possible, provided that the conditions of symmetry are satisfied and that each of the measurements M1, M2, M3, M4 contain information not present in the other measurements.

(129) FIG. 16 illustrates more precisely what is meant by torsion of the plate **58**. As illustrated, two load cells arranged diagonally opposite on the plate **58** measure loads of opposite direction to the other two load cells. Consequently, the plate **58** is subjected to torsion by the action of a couple of opposing forces. The torsion has the effect of creating a saddle-point type of deformation for the

plate **58**. Such torsion can contaminate the weight and offset values, especially when the plate **58** is not sufficiently rigid. Therefore, determining the torsion of the plate **58** can improve the accuracy of the weight and offsets while allowing the possibility of using a relatively flexible plate **58** in the bathroom scale **10**.

(130) To determine the torsion of the plate **58**, the electronic control unit **12** can perform a fourth measurement **M4** of the output signal ΔV with the transistor(s) of the third auxiliary circuit **56** in the closed state. Here, the left front load cell **3AG** no longer contributes to the output signal ΔV . The fourth measurement **M4** is therefore a weight measurement particularly influenced by the right and rear load cells.

(131) Each of measurements **M1** to **M4** corresponds to linear and independent combinations of weight and imbalances. In the control unit **12**, the vector formed by measurements **M1** to **M4** can be multiplied by a transition matrix **A'**. Matrix **A'** here has an additional dimension compared to matrix **A**. Here, matrix **A'** is formed by coefficients **A1** to **A16**. The coefficients can be multiplied by measurements **M1**, **M2**, **M3**, and **M4**. Thus, the fourth measurement **M4** makes it possible to determine, in addition to the weight and offset values, the torsion of the plate **58**.

(132) Note that the torsion can be displayed to the user. The user may for example be asked to correct his position in order to reduce the torsion of the plate **58**. The display of the torsion may be done by displaying indicators **26**, for example by displaying two opposite arrows (on the same diagonal). The user then knows that he must move one foot forward or backward and/or the other foot backward or forward, respectively, to better distribute the loads.

(133) 8. Alternatives to 3 Short Circuits

(134) It appears possible to define alternative positions of the third auxiliary circuit **56**. Each variant can also include alternative circuits with one or two transistors, with or without an additional branch.

(135) However, the following conditions will be satisfied: each of the three auxiliary circuits **42**, **44**, **56** is symmetrical either relative to axis **A1** or to axis **A2**, to avoid amplifier saturation; the three auxiliary circuits **42**, **44**, **56** are arranged so that three load cells out of the four are concerned by an auxiliary circuit.

9. Communication

(136) The electronic control unit **12** may be provided with a communication module **100**. The communication module allows the transmission and/or reception of wireless data.

(137) Preferably, the communication module makes use of a local area network of the Bluetooth, Bluetooth Low-Energy (BLE), or Wi-Fi type. The communication module can then exchange data with a smartphone in proximity to the bathroom scale without requiring significant energy. The smartphone can also serve as a gateway for exchanging data with a remote server. The user can then access the data from the smartphone or from a computer connected to the server.

(138) Alternatively, the communication module makes use of a cellular telecommunications network. The cellular communications network may for example be GSM, 3G, 4G, 5G, 4G-LTE. However, the communication module may also make use of a gateway connected to the cellular network. The gateway may in particular be a router, for example a Wi-Fi router connected to the cellular network. The communication module can then exchange data directly with the server. The user can access the data from the server, in particular on his smartphone or the computer.

(139) The data received by the communication module may comprise information concerning the user, for example name, gender, and a weight history, ideas for questions to ask via the display, and/or the weather forecast for the day.

(140) Furthermore, the communication module can send the weight, a loss of balance, the BCG signal, and/or the answers to the questions asked, to the smartphone and the server. The user can then access the data at any time.

(141) 10. Measurement Method

(142) FIG. **13** illustrates a method, implemented by the electronic control unit **12**, for ascertaining

the weight of a user and his offset on the bathroom scale.

(143) The first step **E1** consists of performing the first measurement **M1** of the output signal ΔV when the four load cells have substantially equal contributions in the Wheatstone bridge. The transistors are then controlled into the open state. Measurement **M1** is a measurement of the weight influenced by the user's right/left and front/rear offsets on the scale.

(144) The second step **E2** consists of controlling the first auxiliary circuit **42** into the closed state in order to short-circuit two resistors belonging to the same foot or two neighboring feet in the bathroom scale.

(145) The third step **E3** consists of performing the second measurement **M2** of the output signal ΔV with the short circuit formed in step **E2**. Measurement **M2** is then a linear combination of the weight with a greater contribution from front/rear or right/left offset.

(146) Note that in the case where the assembly **28** has only one auxiliary circuit, measurement **M1** can be transmitted to the display **14** and measurement **M2** can be used to interact with the bathroom scale and/or to analyze the user's balance and/or to clean the BCG signal in the right/left direction or the front/rear direction.

(147) Where appropriate, step **E4** consists of controlling the first auxiliary circuit **42** into the open state, and controlling the second auxiliary circuit **44** into the closed state. Two resistors belonging to the same foot or to two neighboring feet in the bathroom scale are short-circuited in turn.

Furthermore, a resistor short-circuited by the first auxiliary circuit and at least one resistor short-circuited by the second auxiliary circuit belong to two neighboring feet in the bathroom scale.

(148) The fifth step **E5** consists of performing a third measurement **M3** of the output signal with the second auxiliary circuit **44** in the closed state. Measurement **M3** is also a weight measurement with a greater contribution from a front/rear or right/left offset. However, the contributions of the offsets differ from those of the second measurement **M2**.

(149) In step **E6**, the vector **M** formed by measurements **M1**, **M2**, and **M3** can be multiplied by the transition matrix **A** to obtain a weight value **P** from which the influence of an offset has been eliminated, an offset value in the front/rear direction, and an offset value in the right/left direction.

(150) Thus, in step **E7**, the obtained values are used. The weight **P**, from which the offset has been eliminated, can be displayed to the user. The offset values can be used to clean a BCG signal. The offset values can be used to interact with the scale. The offset values can also be used to analyze the user's balance.

(151) Step **E8** consists of sending the weight, the offset, and/or the BCG signal to the smartphone and the server. The user can then access the measured information without being on the bathroom scale, in particular from his smartphone or a computer.

(152) FIG. **17** illustrates another method **200** comprising an additional step. Steps **E1** to **E5** are identical to those described above for method **100**.

(153) Step **E9** consists of performing a fourth measurement **M4** of the output signal ΔV with the third auxiliary circuit **56** in the closed state. Measurement **M4** is then a measurement of the weight with a contribution from a front/rear or right/left offset that is different from that of the first and second measurements.

(154) Therefore, in step **E6**, the vector **M** formed by measurements **M1**, **M2**, **M3**, and **M4** defines a four-dimensional space. Vector **M** can be multiplied by the transition matrix **A'**. In this manner, in addition to the weight **P** and the offset values in the front/rear and right/left directions, a value corresponding to the torsion of the plate **58** is obtained. Establishing the torsion of the plate makes it possible to improve the precision of the weight and offset values, which can be calculated while taking torsion into account.

(155) Steps **E7** to **E8** are then identical to those described above for method **100**.

(156) The order in which measurements **M1**, **M2**, **M3**, **M4**, etc. are performed is not important. In one embodiment, the frequency **F** of obtaining measurements is between a few Hz and a few hundred Hz, which means that a measurement is made every $1/F$ seconds. Obtaining all the

measurements therefore generally takes less than 0.5 seconds, or even less than 0.1 seconds or less than 0.01 seconds, which are short time intervals, during which the user remains motionless (there will be no displacement in the distribution of mass during the measurements).

(157) The invention is not limited solely to the examples described above but is capable of numerous variations accessible to those skilled in the art.

(158) For example, it is possible to imagine alternative arrangements of auxiliary circuits making it possible to ascertain a weight, a front-rear offset, and/or a right/left offset.

(159) In addition, it is possible to perform a fifth measurement M5 with two auxiliary circuits **42**, **44** in the closed state. This measurement is possible in particular in the case of auxiliary circuits arranged on two neighboring load cells, or in the case of a two-part auxiliary circuit combined with an auxiliary circuit on a load cell distinct from those partly short-circuited by the two-part auxiliary circuit. The fifth measurement M5 can then be added to vector M to ascertain the values of the weights and offsets. Alternatively, the fifth measurement M5 can be a replacement for another measurement, as long as no information is lost among the various measurements M1, M2, M3, M4, or M5.

(160) In addition, one of the measurements can be made with two auxiliary circuits in the closed state at the same time. In this case, the short circuit formed by the two combined auxiliary circuits is symmetrical relative to axis A1 or A2. The Wheatstone bridge thus remains balanced.

(161) Due to the knowledge of the front/rear, left/right imbalances and even due to the torsion, the weighing device can be used as a controller for an electronic system (either the weighing device itself, as described above for when a selection is to be made, or a computer or a game console).

Claims

1. A weighing device for an electronic bathroom scale comprising four feet, including respectively a left front foot, a right front foot, a left rear foot, and a right rear foot, the left front foot comprising a left front load cell, the right front foot comprising a right front load cell, the left rear foot comprising a left rear load cell, and the right rear foot comprising a right rear load cell, each load cell of the left front, right front, left rear and right rear load cells comprising at least two resistors, the left front, right front, left rear and right rear load cells being combined in a Wheatstone bridge assembly comprising: a first branch and a second branch, the first and second branches being mounted in parallel between a reference voltage and a ground potential, the first and second branches being arranged on either side of a first axis of symmetry passing through the reference voltage and the ground potential; a first intermediate point, on the first branch, and a second intermediate point, on the second branch, the first and second branches comprising a same number of resistors on either side of the first and second intermediate points, respectively, so as to define a second axis of symmetry passing through the first and second intermediate points; and a first auxiliary circuit, configured to selectively short-circuit two resistors (a) belonging to a same load cell of a first foot of the four feet in the weighing device so as to short-circuit only said load cell of the first foot in said Wheatstone bridge assembly or (b) belonging to two neighboring load cells of two neighboring first feet of the four feet in the weighing device so as to short-circuit only part of the at least two resistors of each of said two neighboring cells in said Wheatstone bridge assembly, the first auxiliary circuit being symmetrical relative to the first axis or the second axis, the assembly being controlled by an electronic control unit.

2. The weighing device according to claim 1, wherein the assembly comprises a second auxiliary circuit, the second auxiliary circuit being configured to short-circuit two resistors belonging to a same second foot of the four feet or to two neighboring second feet of the four feet in the weighing device, the second auxiliary circuit being symmetrical relative to the first axis or second axis.

3. The weighing device according to claim 2, wherein the assembly comprises a third auxiliary circuit, the third auxiliary circuit being symmetrical relative to the first axis or second axis, the

first, second, and third auxiliary circuits being configured to concern three load cells among the left front, right front, left rear and right rear load cells.

4. The weighing device according to claim 3, wherein the first, second, and third auxiliary circuits are configured to short-circuit the three load cells of the left front, right front, left rear and right rear load cells.

5. The weighing device according to claim 1, wherein the assembly comprises a second auxiliary circuit, the second auxiliary circuit being configured to short-circuit two resistors belonging to a same second foot of the four feet or to two neighboring second feet of the four feet in the weighing device, the second auxiliary circuit being symmetrical relative to the first axis or second axis, and at least one resistor short-circuited by the first auxiliary circuit and at least one resistor short-circuited by the second auxiliary circuit belong to two neighboring feet of the four feet.

6. The weighing device according to claim 1, wherein the first auxiliary circuit comprises at least one switch controlled by the electronic control unit.

7. The weighing device according to claim 1, wherein the first auxiliary circuit comprises two switches controlled by the electronic control unit.

8. The weighing device according to claim 1, wherein the first auxiliary circuit is connected to at least one among: the reference voltage, the ground potential, the first intermediate point, or the second intermediate point.

9. The weighing device according to claim 1, wherein the first and second intermediate points are connected to a single amplifier so as to obtain an output signal from the assembly.

10. A method implemented in a weighing device according to claim 1, comprising: performing a first measurement of an output signal from the assembly; controlling the first auxiliary circuit into a closed state; performing a second measurement of the output signal from the assembly, and establishing a weight and offset of a user based on the first and second measurements.

11. The method according to claim 10, the assembly of the weighing device comprising a second auxiliary circuit, the second auxiliary circuit being configured to short-circuit two resistors belonging to a same second foot of the four feet or to two neighboring second feet of the four feet in the weighing device, the second auxiliary circuit being symmetrical relative to the first axis or second axis, the method further comprising, before or after the second measurement: controlling the second auxiliary circuit into a closed state; performing a third measurement of the output signal from the assembly, and establishing a weight, a front-rear offset, and a right-left offset of the user, based on the first, second, and third measurements.

12. The method according to claim 11, the assembly of the weighing device comprising a third auxiliary circuit, the third auxiliary circuit being symmetrical relative to the first axis or second axis, the first, second, and third auxiliary circuits being configured to concern three load cells among the left front, right front, left rear and right rear load cells, the method further comprising, before or after any among the first, second, or third measurement: controlling the third auxiliary circuit into a closed state; performing a fourth measurement of the output signal from the assembly; and establishing the weight, front-rear offset, right-left offset of the user and a bending of a plate of the weighing device, based on the first, second, third, and fourth measurements.

13. The method according to claim 10, the assembly of the weighing device comprising a second auxiliary circuit, the second auxiliary circuit being configured to short-circuit two resistors belonging to a same second foot of the four feet or to two neighboring second feet of the four feet in the weighing device, the second auxiliary circuit being symmetrical relative to the first axis or second axis, the method further comprising, before or after any among the first or second measurement: controlling the second auxiliary circuit into a closed state; performing a third measurement of the output signal from the assembly; establishing at least a weight of a user and a bending of a plate of the weighing device, based on the first, second, and third measurements.

14. The method according to claim 10, wherein the weight, front-rear offset, and right-left offset is calculated by multiplying a vector, composed of the first, second, and third measurements by a

transition matrix.

15. The method according to claim 14, wherein the transition matrix is obtained by a theoretical calculation.

16. The method according to claim 14, wherein the transition matrix is obtained experimentally.

17. The method according to claim 10, wherein a front-rear and/or right-left offset allows at least one among: an interaction of the weighing device with a user, a cleaning of a ballistocardiography signal, and/or an analysis of the user's balance on the weighing device.

18. The method according to claim 10, further comprising controlling an electronic system, using the obtained offset.

19. A method comprising controlling an electronic system with the weighing device according to claim 1.

20. An electronic device controller, comprising the weighing device according to claim 1.
